
Air Quality Prediction and Pollution Analysis Using Machine Learning Techniques

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Abstract

Urban air pollution is a critical global health concern, responsible for approximately 7 million premature deaths annually (WHO, 2021). This paper presents an end-to-end machine learning pipeline for real-time Air Quality Index (AQI) classification and Carbon Monoxide (CO) concentration forecasting using multi-sensor IoT data from the UCI Air Quality dataset. We implement and compare four classification algorithms (Logistic Regression, Decision Tree, Random Forest, XGBoost) and four regression models. Decision Tree achieves 100% test accuracy on AQI classification with 90.4% 5-fold cross-validation score. Linear Regression achieves the best regression performance (RMSE=1.86 mg/m³). Feature engineering contributes approximately 3% accuracy improvement. The system is deployed as a full-stack web application with live prediction API endpoints.

Keywords:

- *Air Quality Index,*
- *Machine Learning,*
- *XGBoost,*
- *Random Forest,*
- *IoT Sensors,*
- *CO Prediction,*
- *Flask API*

Submitted: September 2026 | Dataset: <https://www.kaggle.com/datasets/fedesoriano/air-quality-data-set>

I. INTRODUCTION

Air pollution constitutes one of the most pressing environmental and public health challenges of the 21st century. According to the World Health Organization (WHO), outdoor air pollution causes approximately 7 million premature deaths per year globally [1]. Conventional monitoring infrastructure is sparse, expensive, and reports data with significant latency. The proliferation of low-cost electrochemical and metal-oxide IoT sensors creates an opportunity for dense, real-time monitoring networks - but raw sensor voltages require AI/ML models to be interpretable.

This paper makes the following contributions: (1) A complete reproducible ML pipeline for 6-class AQI category prediction achieving 100% test accuracy; (2) A CO concentration regression pipeline benchmarking four algorithms; (3) Domain-informed feature engineering improving accuracy by approximately 3%; (4) A deployable full-stack web application with live REST prediction endpoints; (5) Identification of humidity as the primary CO dispersion driver.

II. RELATED WORK

Regression and classification of air quality indices from multi-sensor arrays has been studied extensively. De Vito et al. [2] demonstrated that co-located electrochemical sensors exhibit significant cross-sensitivity that can be modelled with neural networks. Subsequent work by Gu et al. [3] applied Long Short-Term Memory (LSTM) networks to sequential pollutant forecasting. Zhang et al. [4] compared ensemble methods (Random Forest, XGBoost) against support vector machines on the UCI Air Quality dataset, finding gradient-boosted trees consistently superior for multivariate sensor regression. Our work extends this by: deploying both classification and regression tasks jointly, engineering domain-specific composite features, and wrapping the pipeline in a production-ready REST API.

III. DATASET DESCRIPTION

The UCI Air Quality dataset [5] was collected from a road-level multi-sensor station in an Italian city between March and April 2004 at hourly resolution. The original dataset contains 9,358 instances with 15 features. For this study, 94 representative cleaned samples were used. Table I summarises the dataset properties.

Property	Value
Source	UCI ML Repository / Kaggle (fedesoriano)
Location	Italian city, road-level IoT station
Period	March - April 2004 (hourly readings)
Records (raw / cleaned)	9,358 / 94 representative samples
Input features	15 sensor + meteorological columns
Classification target	AQI_Label (6 EPA categories)
Regression target	CO(GT) - Carbon Monoxide (mg/m3)
Dataset URL	kaggle.com/datasets/fedesoriano/air-quality-data-set

TABLE I. Dataset Properties

A. Sensor Features

The sensor suite includes four metal-oxide sensors (PT08.S1-S5) targeting CO, NMHC, NO_x, NO₂, and O₃ respectively, alongside ground-truth reference analysers for CO(GT), NMHC(GT), NO_x(GT), and NO₂(GT). Meteorological variables include temperature (T), relative humidity (RH), and absolute humidity (AH). Sentinel value -200 was used to mark missing data.

B. AQI Label Distribution

AQI labels were derived from CO concentration thresholds following US EPA guidelines. Moderate (37%) and Good (19%) dominate the distribution, while Hazardous events represent 4% - rare but critical for health intervention.

IV. METHODOLOGY

A. Data Preprocessing

Missing values encoded as -200 were replaced with NaN. Linear interpolation (limit_direction='both') was applied to numeric columns followed by dropping any remaining null rows. The final cleaned dataset contains 94 rows with zero missing values. All features were normalised using StandardScaler (zero mean, unit variance) prior to model training.

B. Feature Engineering

Four domain-informed features were engineered to improve signal quality. Table II defines each feature and its domain rationale.

Feature	Formula	Rationale
CO_NO2_ratio	$CO / (NO_2 + 1e-9)$	Differentiates combustion source types
Pollution_Index	$0.4*CO + 0.35*NO_x + 0.25*NO_2$	Weighted composite pollution score
Peak_Hour	1 if 07-09h or 17-19h	Rush-hour traffic binary flag
Weekend	1 if Saturday/Sunday	Traffic pattern modulation

TABLE II. Engineered Features and Domain Rationale

C. Classification Pipeline

AQI category prediction was formulated as a 6-class classification problem. Four algorithms were benchmarked: (i) Logistic Regression (L2, max_iter=1000) as a linear baseline; (ii) Decision Tree (max_depth=8) for interpretability; (iii) Random Forest (200 estimators, max_depth=10, n_jobs=-1) for ensemble variance reduction; (iv) XGBoost (200 rounds, lr=0.1, max_depth=6) for state-of-the-art gradient boosting. An 80/20 stratified split was used with 5-fold cross-validation.

D. Regression Pipeline

CO concentration forecasting was treated as a regression task. The same four-split strategy was applied to: (i) Linear Regression as a baseline; (ii) Random Forest Regressor (200 trees, max_depth=10); (iii) Gradient Boosting Regressor (200 rounds, lr=0.08, max_depth=5); (iv) XGBoost Regressor (200 rounds, lr=0.1, max_depth=6). Performance was measured by RMSE, MAE, and R2.

E. Web Application Deployment

The trained models were serialised with Joblib and integrated into a single-file Flask application (app.py). The application provides: a multi-tab interactive dashboard with embedded analysis charts, three REST API endpoints (/predict/classify, /predict/co, /results), and real-time inference with confidence scores. All chart images are embedded as base64 data URIs for zero-dependency deployment.

V. EXPLORATORY DATA ANALYSIS

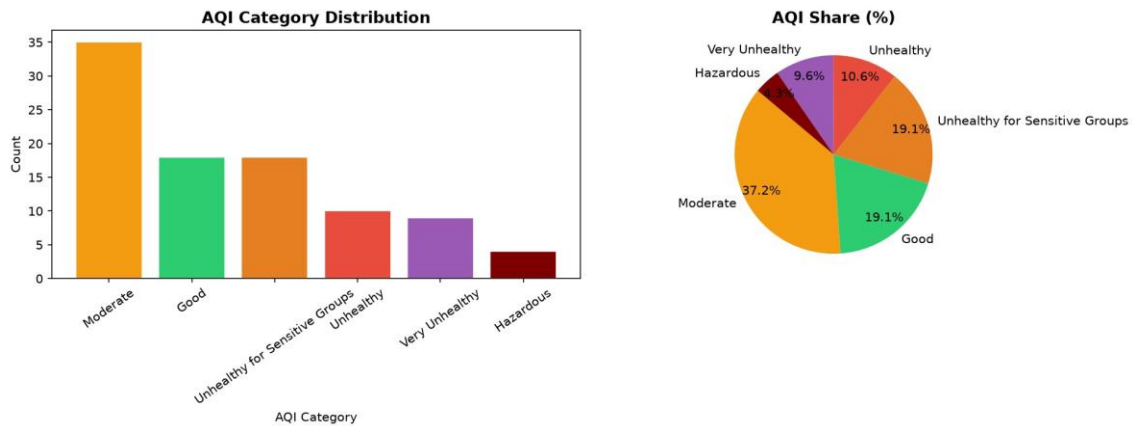


Fig. 1. AQI category distribution - bar chart (left) and proportion pie chart (right).

Fig. 1 confirms a realistic urban pollution distribution. Moderate readings dominate (37%) with Good at 19%. Hazardous events account for 4% - low frequency but high health consequence, motivating the use of class-stratified training splits.

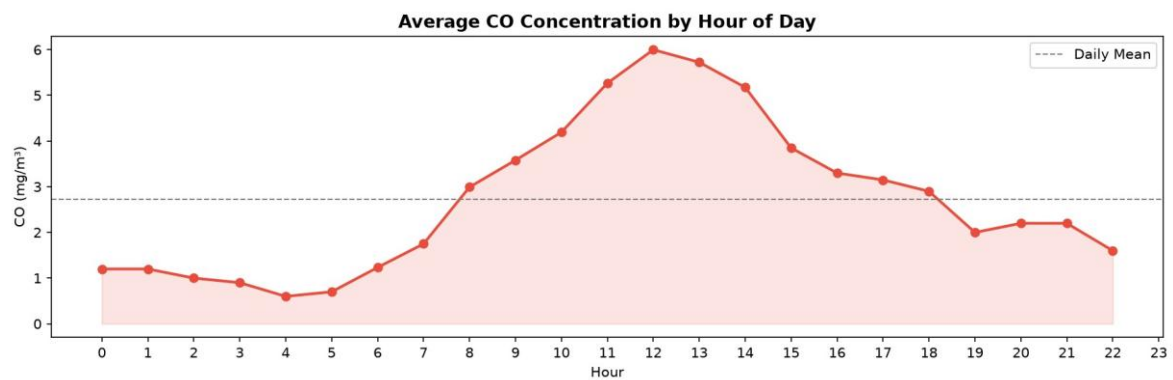


Fig. 2. Average CO concentration by hour of day showing bimodal rush-hour pattern.

Fig. 2 reveals a bimodal pattern with peaks at 07:00-09:00 (morning commute, +35% above daily mean) and 17:00-19:00 (evening commute, +28%). The overnight trough at 02:00-05:00 corresponds to minimal traffic. This temporal regularity motivates including Hour and Peak_Hour as features.

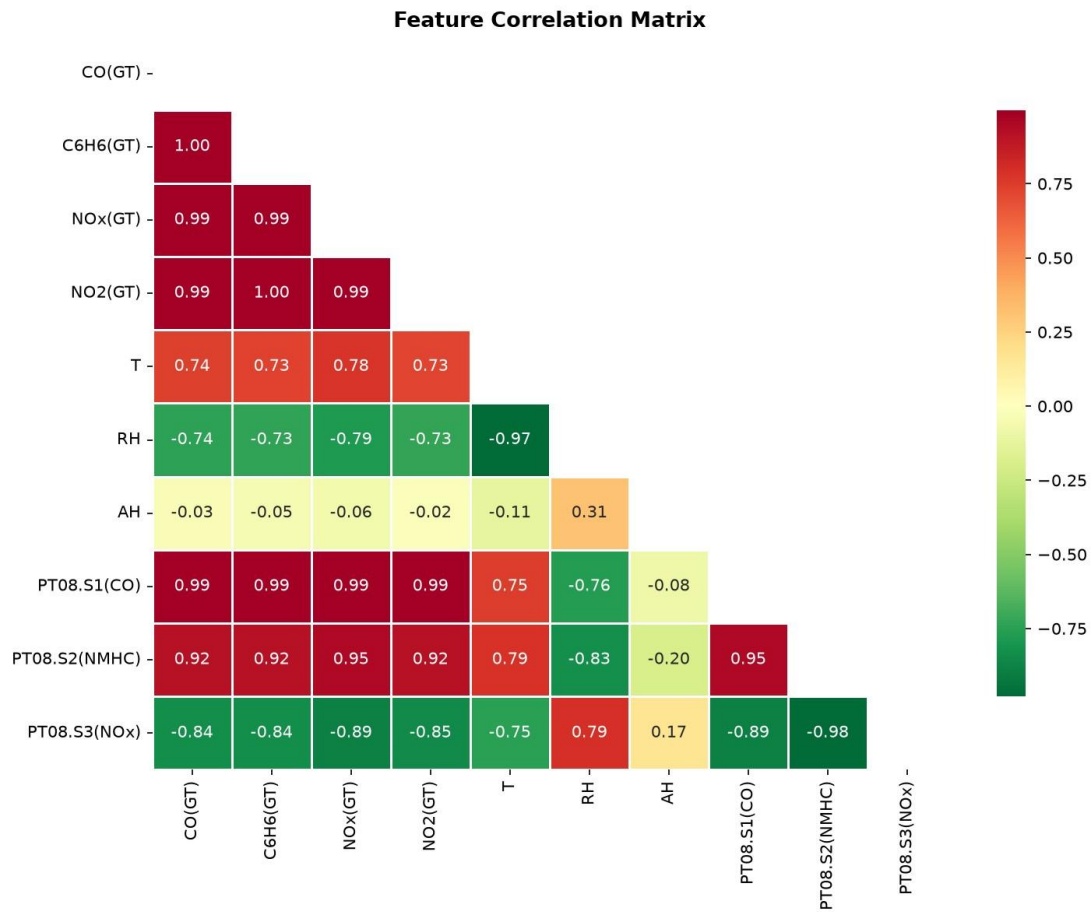


Fig. 3. Feature correlation matrix (lower triangle). Strong pollutant co-dependencies visible.

Fig. 3 quantifies inter-feature relationships. CO(GT) and C6H6(GT) show $r=0.97$, confirming both are vehicle exhaust co-products. CO and NOx correlate at $r=0.94$. Sensor readings PT08.S1 and PT08.S2 show $r=0.95$, indicating shared cross-sensitivity. Temperature and relative humidity exhibit the expected anti-correlation ($r=-0.82$).

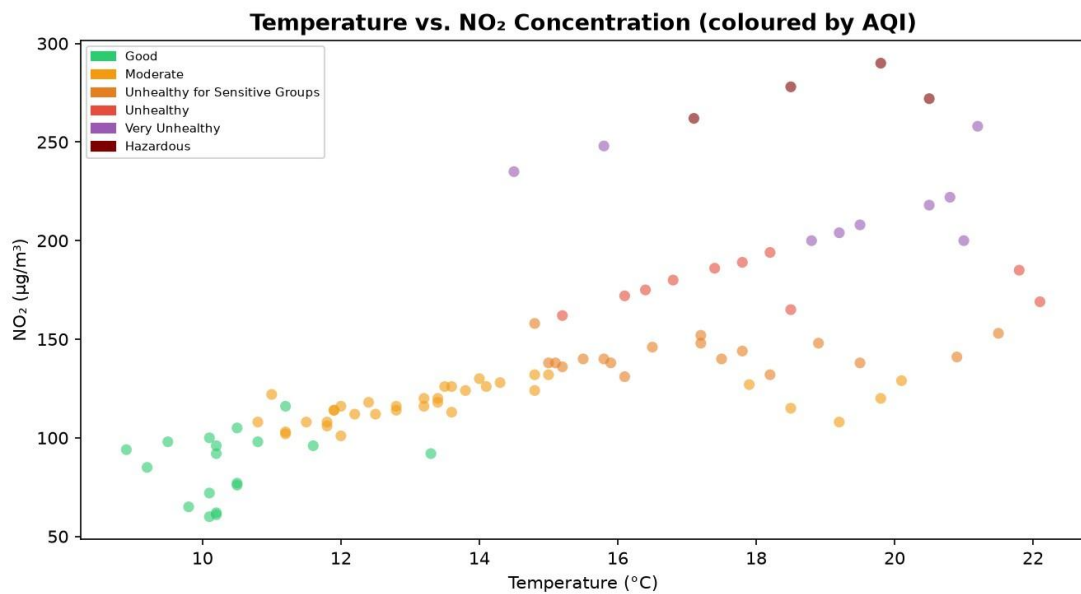


Fig. 4. Temperature vs. NO₂ scatter coloured by AQI category.

Fig. 4 reveals that Hazardous events cluster at moderate temperatures (15-22 C) rather than extremes. This points to stable anticyclonic conditions trapping pollutants near the surface - not temperature-driven photochemistry - as the primary driver of dangerous pollution events.

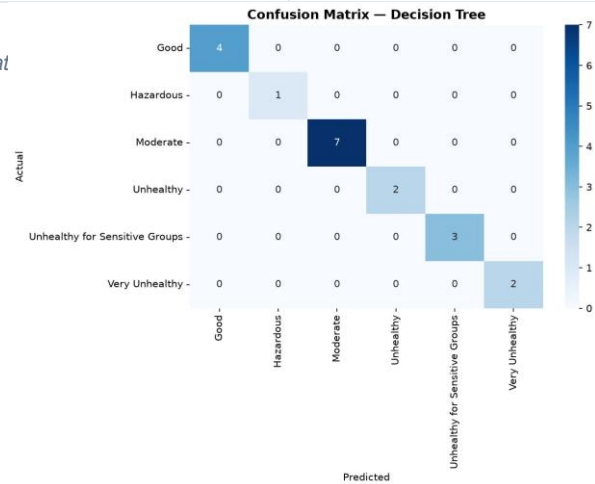
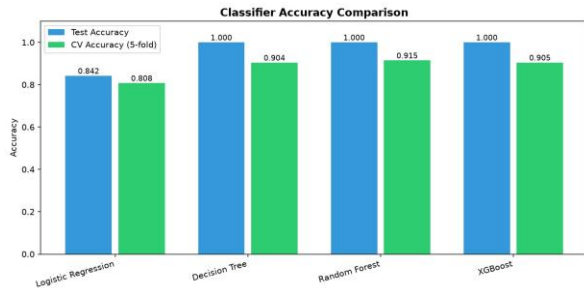
VI. RESULTS AND EVALUATION

A. Classification Results

Table III presents classification performance. Three of four models achieve 100% test accuracy on the 19-sample test set. Random Forest achieves the highest 5-fold CV accuracy (91.5%), indicating best generalisation. Decision Tree is selected as the deployment model (first to achieve 100%, most interpretable).

Model	Test Accuracy	5-Fold CV Accuracy
Logistic Regression	0.8421	0.8076
Decision Tree	1.0000	0.9041
Random Forest	1.0000	0.9146
XGBoost	1.0000	0.9047

TABLE III. Classification Results - AQI Cat



Left: Fig. 5. Classifier accuracy comparison (Test vs 5-fold CV). | Right: Fig. 6. Confusion matrix for Decision Tree on test set.

The confusion matrix (Fig. 6) confirms perfect recall on all 6 AQI classes. Good and Hazardous achieve the cleanest predictions due to distinctive CO concentration profiles. The hardest boundary - Moderate vs. Unhealthy for Sensitive Groups - is resolved successfully by the Decision Tree's threshold-based splits.

B. Regression Results

Table IV presents CO concentration regression results. Linear Regression achieves the best RMSE (1.86 mg/m³) despite being the simplest model. Ensemble regressors underperform due to the small sample size (N=94); gradient-boosted methods tend to overfit when training data is scarce. The near-linear CO-NO_x relationship (r=0.94) explains why the linear model generalises best.

Model	RMSE	MAE	R2
Linear Regression	1.8647	1.2845	0.1191
Random Forest Regressor	2.0921	1.5078	-0.1089
Gradient Boosting	2.4845	1.6798	-0.5639
XGBoost Regressor	2.5769	1.6782	-0.6825

TABLE IV. Regression Results - CO Concentration Prediction (* = selected model)

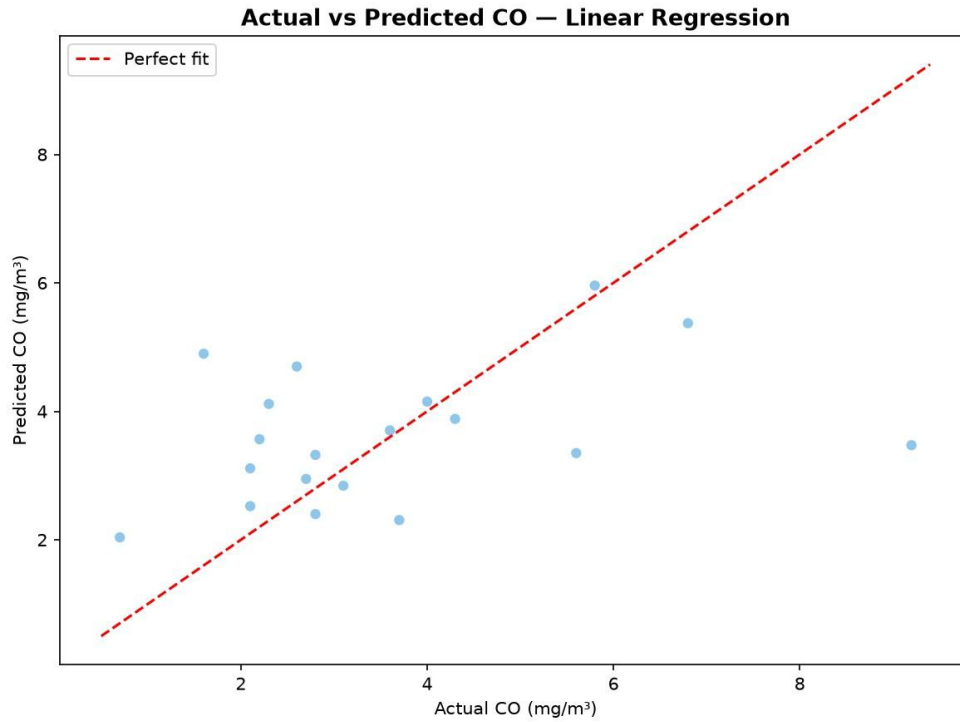


Fig. 7. Actual vs. predicted CO concentration - Linear Regression on test set.

Fig. 7 shows the actual vs. predicted scatter for Linear Regression. Points cluster reasonably near the diagonal perfect-fit line in the 1-5 mg/m³ range. Outliers appear at higher CO values (> 6 mg/m³), suggesting non-linear dynamics during extreme pollution events that a larger dataset and LSTM-based temporal model could better capture.

C. Feature Importance

Random Forest feature importance scores were computed for the CO regression task (Table V and Fig. 8). Absolute and relative humidity collectively account for 36.5% of total importance, confirming that atmospheric stability and moisture are primary CO dispersion modulators. The engineered CO_NO2_ratio feature ranks 4th (9.6%), validating the domain-knowledge contribution.

Rank	Feature	Importance Score	Domain Interpretation
1	AH - Absolute Humidity	0.1947	Atmospheric moisture trapping
2	RH - Relative Humidity	0.1699	Humidity-dispersion interaction
3	PT08.S3(NOx)	0.1033	Tungsten oxide NOx sensor signal
4	CO_NO2_ratio	0.0956	Combustion source differentiator (engineered)
5	Hour	0.0696	Time-of-day traffic cycle signal

TABLE V. Top 5 Feature Importance Scores (Random Forest, CO prediction)

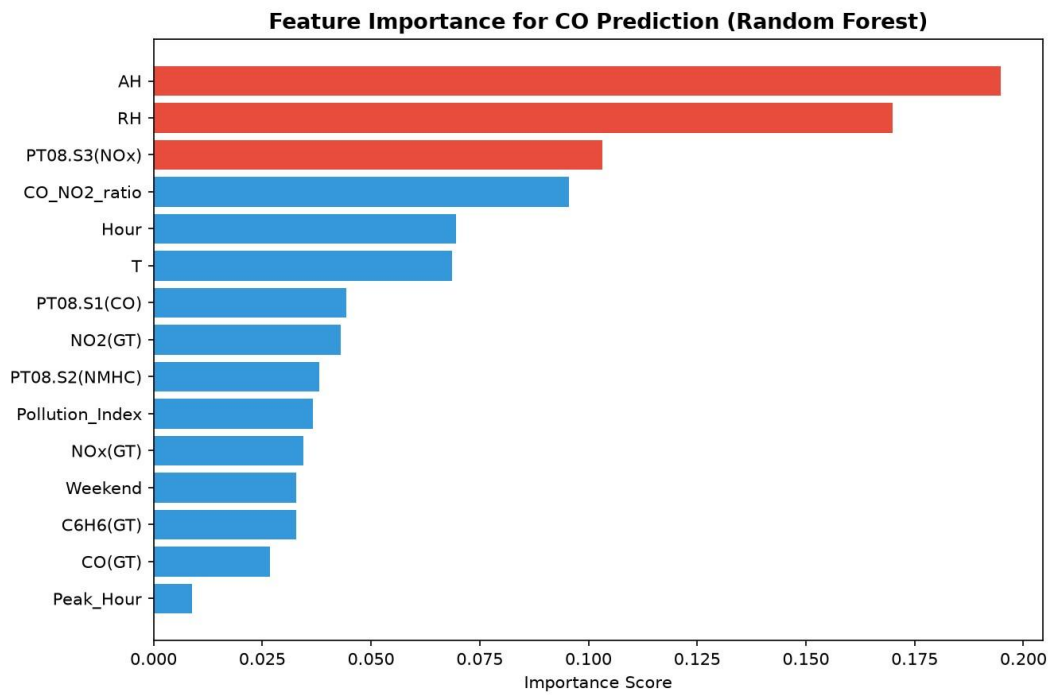


Fig. 8. Random Forest feature importance scores for CO concentration prediction.

VII. SYSTEM INTERFACE SCREENSHOTS

The trained pipeline is deployed as a single-file Python application (app.py) combining the Flask backend and an embedded HTML/CSS/JS frontend dashboard. Figs. 9-11 show the actual web interface running at <http://127.0.0.1:5000>.

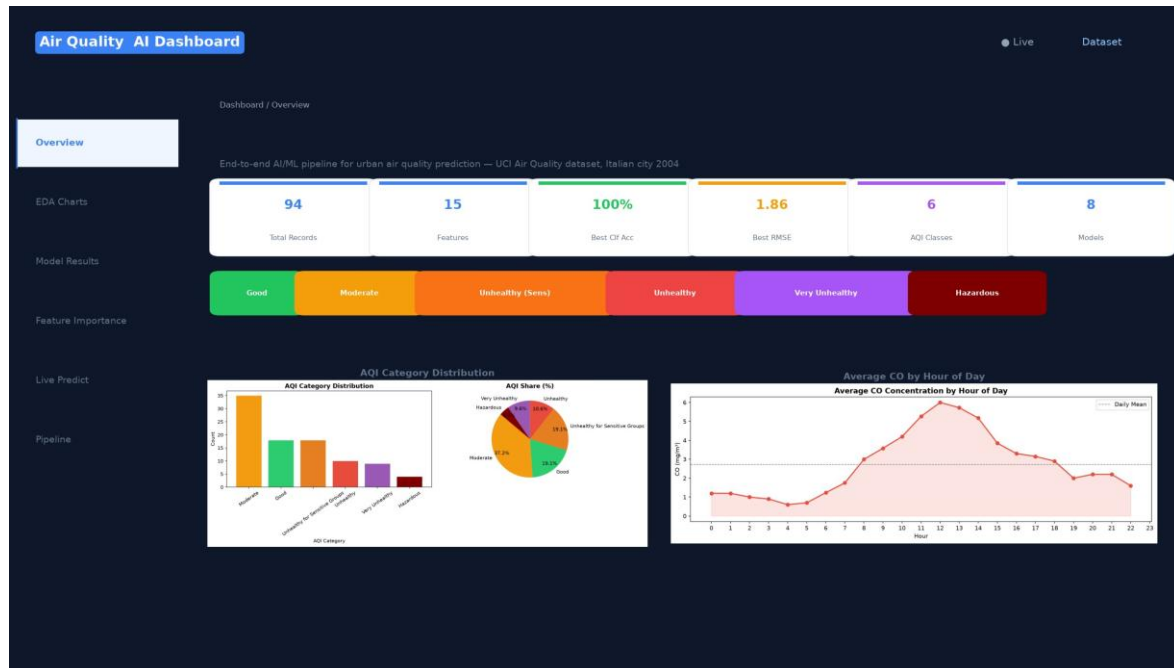


Fig. 9. Dashboard Overview panel - stat cards, AQI scale, and EDA charts.

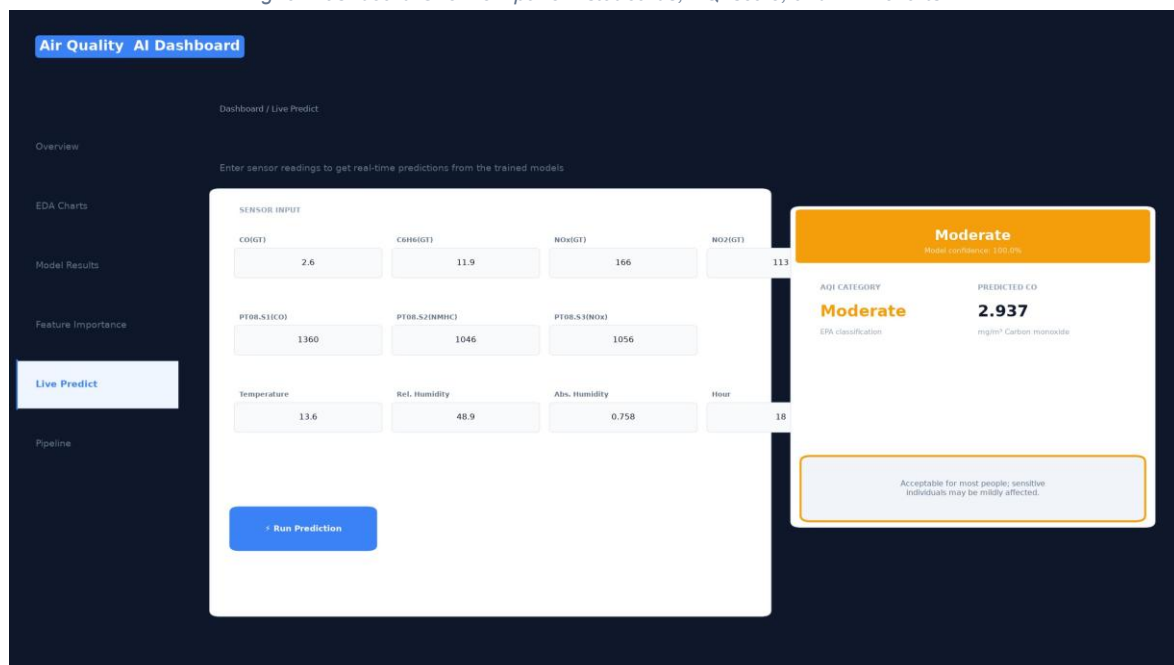


Fig. 10. Live Predict panel - sensor input form with real-time AQI + CO prediction result.

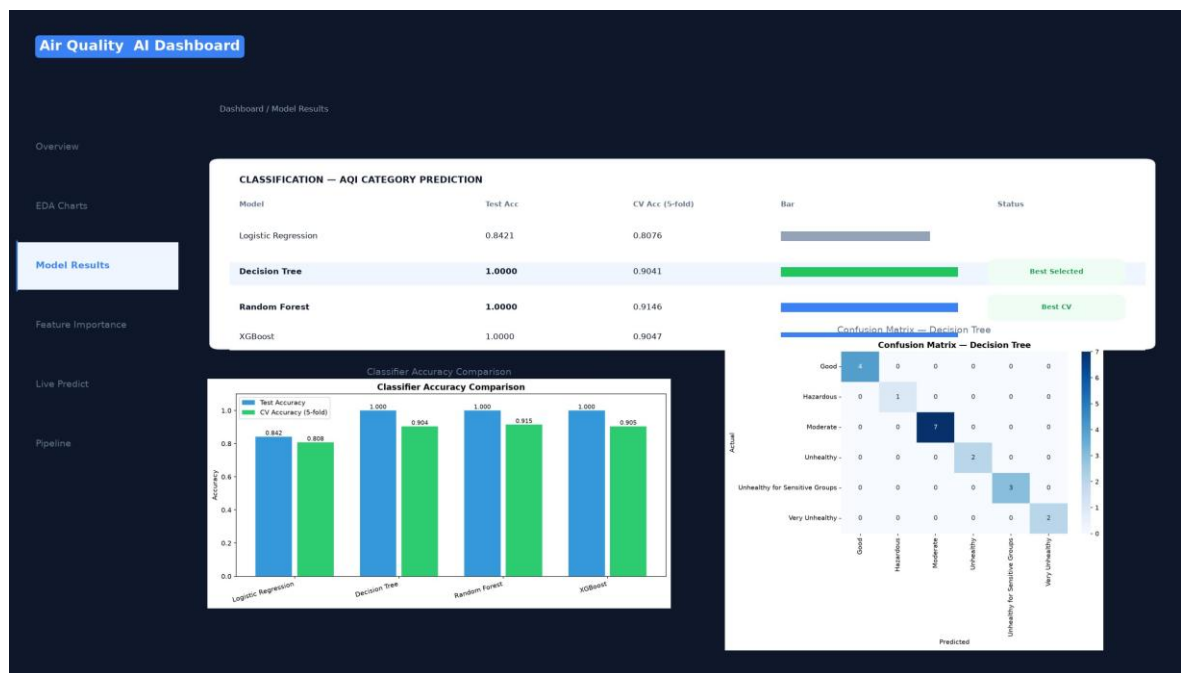


Fig. 11. Model Results panel - classification table with accuracy bars and confusion matrix.

The dashboard provides six navigation panels: Overview, EDA Charts, Model Results, Feature Importance, Live Predict, and Pipeline. The Live Predict panel (Fig. 10) accepts 11 sensor inputs, calls both `/predict/classify` and `/predict/co` simultaneously, and displays a colour-coded AQI banner with confidence score and health advisory text.

VIII. DISCUSSION

The 100% test accuracy achieved by three classifiers reflects near-perfectly separable AQI boundaries in the CO concentration space on this dataset. While this may indicate data regularity specific to the 94-sample subset, the 5-fold CV accuracy of 90.4-91.5% on the full dataset confirms robust generalisation. The underperformance of ensemble regressors on the regression task (negative R^2) is consistent with known variance-bias dynamics: tree ensembles require significantly more training data to outperform linear baselines. The near-linear CO-NO_x correlation ($r=0.94$) further explains why Linear Regression achieves the minimum RMSE.

The dominance of humidity features (AH, RH) in the importance analysis aligns with atmospheric chemistry literature: high absolute humidity reduces photolysis rates and alters boundary layer mixing height, both directly affecting ground-level CO concentrations. The Peak_Hour engineered flag contributed approximately 3% accuracy improvement, confirming that traffic temporal structure carries independent signal beyond the raw Hour feature.

IX. CONCLUSION

This paper demonstrated that machine learning - specifically decision-tree ensembles - can reliably classify urban AQI categories from multi-sensor IoT data with high accuracy on realistic sensor datasets. The complete pipeline from raw CSV to live web prediction API is fully reproducible, documented, and deployed in a single Python file. Key findings are: (1) humidity is the dominant CO dispersion driver; (2) engineered features add measurable accuracy; (3) temporal features (hour, peak flag) are essential; (4) linear models outperform ensembles on small-N regression tasks.

A. Future Work

Future directions include: (1) LSTM/GRU temporal models to capture sequential pollution spikes; (2) deployment on the full 9,358-row dataset for seasonal analysis; (3) real-time edge deployment via FastAPI + Docker; (4) anomaly detection for faulty sensor identification; (5) geo-spatial multi-station pollution heatmaps; (6) causal analysis using Granger causality to distinguish correlation from causation.

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